arXiv:2511.08152v1 [cs.CV] 11 Nov 2025

Boomda: Balanced Multi-objective Optimization for Multimodal Domain Adaptation

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Abstract

Multimodal learning, while contributing to numerous success stories across various fields, faces the challenge of prohibitively expensive manual annotation. To address the scarcity of annotated data, a popular solution is unsupervised domain adaptation, which has been extensively studied in unimodal settings yet remains less explored in multimodal settings. In this paper, we investigate heterogeneous multimodal domain adaptation, where the primary challenge is the varying domain shifts of different modalities from the source to the target domain. We first introduce the information bottleneck method to learn representations for each modality independently, and then match the source and target domains in the representation space with correlation alignment. To balance the domain alignment of all modalities, we formulate the problem as a multi-objective task, aiming for a Pareto optimal solution. By exploiting the properties specific to our model, the problem can be simplified to a quadratic programming problem. Further approximation yields a closed-form solution, leading to an efficient modality-balanced multimodal domain adaptation algorithm. The proposed method features **B**alanced multi-**o**bjective **o**ptimization for **m**ultimodal **d**omain **a**daptation, termed **Boomda**. Extensive empirical results showcase the effectiveness of the proposed approach and demonstrate that Boomda outperforms the competing schemes. The code is is available at: https://github.com/sunjunaimer/Boomda.git.

1 Introduction

Multimodal learning typically leverages heterogeneous and complementary signals, such as acoustic, visual, and linguistic information, to perform machine learning tasks like classification, clustering and retrieval. Thanks to recent advances in hardware and model design, multimodal learning has been applied to various applications, including but not limited to action recognition (Woo et al. 2023), affective computing (Sun et al. 2023; Zhang et al. 2024), and medical analysis (Liu et al. 2023). Compared with the unimodal alternative, multimodal learning achieves remarkable performance improvement. However, one of its notorious drawbacks is that collecting and annotating multimodal data is expensive and time-consuming. Consequently, the scarcity of annotated data presents a major challenge for the practical application of multimodal learning.

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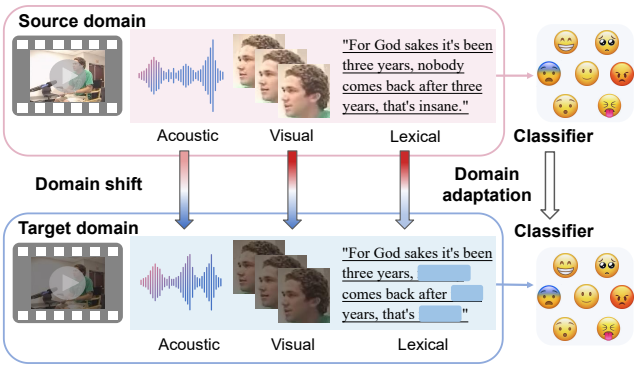

Figure 1: Domain shift in the context of multimodal emotion recognition. The example sample is drawn from dataset IEMOCAP (Busso et al. 2008).

To alleviate this issue, unsupervised domain adaptation has merged as a powerful approach for enhancing the model's ability to transfer knowledge from a label-rich source domain to an unlabeled but related target domain. The objective of unsupervised domain adaptation is to train a model on the source domain while optimizing its performance on the target domain, which lacks labels during training. Mainstream approaches usually align source and target domains and learn generalizable representations via either directly minimizing the feature distribution shift between the two domains, or training an adversarial domain classifier. A massive amount of techniques following these principles have been developed, demonstrating impressive results across various tasks, including image classification (Hu et al. 2023; Lee et al. 2023), semantic segmentation (Wang et al. 2023), object detection (Kennerley et al. 2023), question answering (Dua et al. 2023), among others.

Nevertheless, the existing domain adaptation literature primarily focuses on unimodal settings, particularly within the areas of computer vision and natural language processing. In comparison, multimodal domain adaptation remains less investigated, yet is attracting growing interests due to the popularity of multimodal learning (Sun et al. 2025). From the perspective of modality diversity, multimodal domain adaptation is conceptually divided into two cate-